

Unsupervised Learning - Lab 09 - Bayesian Networks

Parameter Learning

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Agenda

1. Load the Alarm model from file.
2. Visualize the model structure.
3. Generate data from the model.
4. Learn the parameters using Maximum Likelihood Estimator.
5. Learn the parameters using the Bayesian Estimator.
6. Learn the parameters in presence of missing values.

Introduction to Bayesian Networks

- Bayesian Networks are probabilistic graphical models that represent a set of variables and their conditional dependencies via a Directed Acyclic Graph (DAG).
- Today, we will use the `pgmpy` library.

Maximum Likelihood Estimator

- In case of categorical values, the Maximum Likelihood Estimator is given by the **counts of the categories**.

- Dataset

	rain
1	yes
2	no
3	no

- Counts

rain	no	yes	total
	2	1	3

- Probability

rain	no	yes	total
	67%	33%	100%

Bayesian Estimator

- In case of categorical values, the Bayesian Estimator is given by the counts of the categories **plus some prior knowledge**.

- **Pseudo counts** can be used to represent our prior knowledge.

rain	no	yes	total
	5	2	7

- We can also use probabilities values directly and normalize them.

- Dataset

	rain
1	yes
2	no
3	no

- Counts

rain	no	yes	total
	2 + 5	1 + 2	3 + 7

- Probability - $P(\text{rain})$

rain	no	yes	total
	70%	30%	100%

Estimating in Presence of Missing Values

- When some values are missing there are **various strategies** that can be used to deal with them.
- If we have enough values, we can **delete them**.

- Dataset

	rain
1	yes
2	?
3	no

- Counts

rain	no	yes	total
	1	1	2

- Probability - $P(\text{rain})$

rain	no	yes	total
	50%	50%	100%

Conditional Probability Estimation

- What about **conditioning**?
- We estimate **conditional counts**.

- Dataset

	rain	wind
1	yes	yes
2	no	no
3	no	yes

- Counts

rain	no	yes	total
wind = no	1	0	1
wind = yes	1	1	2

- Probability - $P(\text{rain} \mid \text{wind})$

rain	no	yes	total
wind = no	100%	0%	100%
wind = yes	50%	50%	100%

- pgmpy documentation: pgmpy.org
- Alarm network BIF file: [BN Repository](#)